

Markov inequality: $X \geq 0$

$$P(X \geq u) \leq \frac{E(X)}{u}$$

$$E(X) = \int_{-\infty}^u u f(u) du + \int_u^{\infty} u f(u) du.$$

$$\begin{aligned} &\geq \int_u^{\infty} u f(u) du \\ &\geq u \int_u^{\infty} f(u) du \\ &= u P(X \geq u) \end{aligned} \quad \left. \begin{array}{l} \text{on } \mathbb{R} \text{ mean } \mu, \text{ var } \sigma^2 \\ P(|\bar{X} - \mu| \geq \epsilon) \leq \frac{\sigma^2}{\epsilon^2} \\ \text{for any } \epsilon > 0 \end{array} \right\}$$

Choose $X = (\bar{X} - \mu)^2$, $u = \epsilon^2$

$$\frac{E[(\bar{X} - \mu)^2]}{\epsilon^2} \geq P((\bar{X} - \mu)^2 \geq \epsilon^2)$$

$$\Rightarrow P(|\bar{X} - \mu| \geq \epsilon) \leq \frac{\sigma^2}{\epsilon^2} \quad \text{for any } \epsilon > 0$$

① $X \sim \text{exp}$ with mean $E(X) = 1$

$u = 0.5$
Compare $P(X > u)$ with Markov inequality bound $\frac{E(X)}{u}$

$$P(X > u) = e^{-u}$$

$$P(X > \frac{1}{2}) = e^{-1/2}$$

$$\frac{E(X)}{u} = \frac{1}{1/2} = 2$$

② $X \sim \text{Bernoulli}(p)$

Compare $P(X > 1)$ with Markov inequality bound $E(X)$

Wichtig: $\frac{1}{\mu}$ ist die mittlere Teilchenlänge $\frac{1}{\mu}$

$$P(X \geq 1) = P(X = 1) = p \quad ; \quad \underline{E(X)} = p$$

③ Let $U_1, U_2, \dots$ be indep. r.v.'s each uniformly distributed over the interval $[0, 1]$. Show that

$$X_0 = 1 \text{ and } X_n = 2^n U_1 \dots U_n \text{ for } n = 1, 2, \dots \text{ in a MG}$$

Sied

$$X_{n+1} = 2 X_n U_{n+1}$$

$$E(X_{n+1} | X_0, \dots, X_n) = E(2X_n U_{n+1} | X_0, \dots, X_n)$$

$$= 2 X_n E(U_{n+1} | U_0, \dots, U_n)$$

$$\begin{aligned} & \stackrel{\text{ind. of } U_1, \dots, U_n}{=} 2X_n \underbrace{E(U_{n+1})} \rightarrow \int_0^1 u \cdot 1 du = \frac{1}{2} \\ & = 2X_n \cdot \frac{1}{2} \end{aligned}$$

$$14 \times 2$$

$$(X_n) \text{ is a martingale}$$

4) Let $S_0 = 0$ for $n \geq 1$, $S_n = \epsilon_1 + \dots + \epsilon_n$
be sum of n indep. var., each expo. distributed
with mean $E(\epsilon) = 1$. $X_n = 2^n e^{-S_n}$, $n \geq 0$
is mgf?

Sred

$$X_{n+1} = 2 X_n e^{-\epsilon_{n+1}}$$

$$E(X_{n+1} | X_0, \dots, X_n) = E(2X_n e^{-\epsilon_{n+1}} | X_0, \dots, X_n)$$

$$= 2 X_n \mathbb{E}(e^{-\epsilon_{n+1}} | \epsilon_0, \dots, \epsilon_n)$$

$$\stackrel{\text{ind.}}{=} 2X_n E(e^{-G_{n+1}}) \quad \int_0^\infty e^{-u} du = 1$$

$$\frac{1}{0} \leq \frac{1}{0} \leq \frac{1}{0} = \frac{1}{0}$$

$$= 2X_n \cdot \frac{1}{2} = 1$$

⑤ Let $\varepsilon_1, \varepsilon_2, \dots$ be indep Bernoulli's with parameter p , $0 < p < 1$, show that $X_0 = 1$ and $X_n = p^{-n} \varepsilon_1 \dots \varepsilon_n$, $n = 1, 2, \dots$ is MG

What is the limit X_n as $n \rightarrow \infty$.

sol

$$X_{n+1} = p^{-1} \varepsilon_{n+1} X_n$$

$$E(X_{n+1} | X_0, X_1, \dots, X_n) = E(p^{-1} \varepsilon_{n+1} X_n | X_0, \dots, X_n)$$

$$= X_n E(p^{-1} \varepsilon_{n+1} | X_0, \dots, X_n)$$

$$= X_n E(p^{-1} \varepsilon_{n+1} | \varepsilon_0, \varepsilon_1, \dots, \varepsilon_n)$$

$$\text{by independence of } \varepsilon_i\text{'s} = X_n E(p^{-1} \varepsilon_{n+1}) = X_n \times p^{-1} \times E(\varepsilon_{n+1}) = \underline{\underline{X_n}}$$

$$\varepsilon_i = \begin{cases} 1 & \text{w.p. } p \\ 0 & \text{w.p. } 1-p \end{cases}$$

$$\prod_{i=1}^n \varepsilon_i = \begin{cases} 1 & \text{w.p. } p^n \\ 0 & \text{w.p. } 1-p^n \end{cases}$$

$$X_n = \begin{cases} \frac{1}{p^n} & \text{w.p. } p^n \\ 0 & \text{w.p. } 1-p^n \end{cases}$$

$$n \rightarrow \infty$$

$$X_n \Rightarrow 0 \text{ w.p. } 1$$

⑥ Consider a SP that evolves according to the following laws: If $X_n = 0$, then $X_{n+1} = 0$, whereas if $X_n > 0$, then

$$X_{n+1} = \begin{cases} X_n + 1 & \text{wp } \frac{1}{2} \\ X_n - 1 & \text{wp } \frac{1}{2} \end{cases}$$

sol X_n is MG.

$$E(X_{n+1} | X_0 = x_0, \dots, X_n = x_n)$$

$$= (x_n + 1) \times \frac{1}{2} + (x_n - 1) \times \frac{1}{2}$$

$$= x_n$$

$$E(X_{n+1} | X_0, \dots, X_n) = X_n$$

$\therefore X_n$ is MG.